

Module 1.1: Introduction to Big Data

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Big Data

Big Data refers to extremely large, complex, and rapidly growing datasets that cannot be effectively stored, processed, managed, or analyzed using traditional database systems and software tools.

Traditional databases are designed to handle structured data of limited size. However, with the growth of the internet, social media, mobile devices, cloud computing, and IoT, enormous amounts of data are generated every second. This massive volume of data requires specialized technologies and techniques for storage, processing, and analysis.

Sources of Big Data

- Social media platforms (Facebook, Instagram, X)
- E-commerce websites (Amazon, Flipkart)
- Search engines (Google)
- Banking transactions
- Healthcare systems
- IoT devices and sensors
- Mobile applications
- Scientific research

Examples of Big Data

- YouTube receives hundreds of hours of video uploads every minute.
- Facebook users generate millions of posts, comments, and likes daily.
- Online shopping platforms record millions of customer transactions.
- Hospitals store huge volumes of patient records and medical images

Characteristics of Big Data (The 5 Vs)

Big Data is commonly described using five important characteristics known as the **5 Vs**.

1. **Volume:** Volume refers to the enormous quantity of data generated and stored. As organizations collect information from various sources, the amount of data grows rapidly, reaching terabytes, petabytes, and even exabytes. **Examples**

- Social media posts
- Online transactions
- Video streaming data
- Sensor-generated data

Importance: Large volumes of data require scalable storage systems and distributed processing techniques. **Example:** A large e-commerce company may store several petabytes of customer transaction data.

2. **Velocity:** Velocity refers to the speed at which data is generated, transmitted, and processed. Modern applications often require real-time or near real-time analysis. **Examples**

- Stock market transactions
- Credit card payments
- GPS tracking systems
- Social media updates

Importance: Organizations must process data quickly to make timely decisions. **Example:** Banks analyze credit card transactions instantly to detect fraud.

3. **Variety:** Variety refers to the different types and formats of data. Unlike traditional databases that mainly handle structured data, Big Data includes multiple forms of information. **Types of Data**

a) **Structured Data:** Data organized in rows and columns. Examples:

- Employee databases
- Student records

b) **Semi-Structured Data:** Data that contains tags or markers but does not follow a rigid structure. Examples:

- XML
- JSON

c) **Unstructured Data:** Data without a predefined format. Examples:

- Images
- Videos
- Audio files
- Emails
- Social media posts

Importance

Organizations must manage and analyze data from multiple sources and formats.

4. **Veracity:** Veracity refers to the quality, reliability, and accuracy of data. Not all collected data is correct or trustworthy. Poor-quality data can lead to incorrect conclusions. **Problems Affecting Veracity**

- Missing values
- Duplicate records
- Inconsistent information
- Fake or misleading data

Importance: Reliable data improves decision-making and analytical accuracy. **Example:** Fake social media accounts may generate inaccurate information that affects business analysis

5. **Value:** Value refers to the useful insights and benefits extracted from data. Collecting large amounts of data is meaningless unless organizations can derive valuable information from it.

6. **Importance:** Big Data helps organizations:

- Improve decision-making
- Increase profits

- Understand customers
- Optimize operations

Example: Netflix analyzes user viewing habits to recommend movies and TV shows.

	Meaning
Volume	Amount of data
Velocity	Speed of data generation and processing
Variety	Different forms of data
Veracity	Accuracy and reliability of data
Value	Useful insights obtained from data



Why Big Data?

Traditional database systems were designed to manage relatively small amounts of structured data. As technology advanced, data began growing at an unprecedented rate. Conventional systems became insufficient for handling this massive volume and complexity.

Big Data technologies were developed to overcome these limitations.

Reasons for the Emergence of Big Data

- **Explosion of Digital Data:** The rapid growth of digital devices and online services continuously generates huge amounts of data from social media, mobile apps, e-commerce platforms, and sensors.
- **Growth of the Internet:** Billions of users interact online daily, creating vast amounts of information through searches, website visits, and online transactions.
- **Rise of social media:** Social networking platforms generate massive volumes of user-created content such as posts, comments, images, and videos.
- **Cloud Computing:** Cloud platforms provide scalable and cost-effective storage and computing resources, making it easier to handle large datasets.
- **Mobile Computing:** Smartphones and mobile applications continuously generate data related to user activities, location, communication, and online services.
- **Internet of Things (IoT):** Connected devices like smart watches, smart homes, and industrial sensors continuously collect and transmit data.
- **Need for Real-Time Analytics:** Organizations require immediate analysis of data for applications such as fraud detection, traffic management, and recommendation systems.
- **Advances in Storage Technology:** Storage has become cheaper and more accessible, enabling organizations to store massive amounts of data.
- **Competitive Business Environment:** Organizations use data-driven strategies to gain competitive advantages, improve decision-making, and enhance customer satisfaction.

Advantages of Big Data

1. **Better Decision Making:** Helps organizations make accurate decisions using historical and real-time data.
2. **Improved Customer Experience:** Enables businesses to understand customer preferences and provide personalized services.
3. **Fraud Detection:** Identifies suspicious activities and helps prevent fraud in real time.

4. **Risk Management:** Assists organizations in identifying and reducing potential risks.
5. **Increased Efficiency:** Optimizes business processes and improves resource utilization.
6. **Innovation:** Supports the development of new products, services, and business strategies.

Limitations and Challenges of Big Data

1. Storage Complexity
2. Data Security Issues
3. Privacy Concerns
4. Data Quality Problems
5. High Processing Requirements
6. Skilled Workforce Requirements

Types of Data

Data can be classified into three main categories based on its structure and organization: **Structured Data, Semi-Structured Data, and Unstructured Data**. Understanding these types is important because Big Data systems are designed to handle all forms of data.

1. **Structured Data:** is data that is organized in a predefined format, usually in rows and columns. It follows a fixed schema, making it easy to store, search, and analyze using traditional systems DBMS.

- Organized in tabular form.
- Follows a predefined schema.
- Easy to store and retrieve.
- Can be managed using SQL databases.
- Highly structured and consistent.

Example Table

Student ID	Name	Marks
101	Amit	85
102	Rahul	90

Advantages

- Easy querying and analysis.
- High data integrity.
- Efficient storage and retrieval.

Limitations

- Difficult to handle complex and changing data formats.
- Not suitable for multimedia content.

Examples

- Student records
- Employee databases
- Banking transactions
- Sales records

2. **Semi-Structured Data:** does not follow a rigid table structure but contains tags, labels, or markers that help organize and describe the data. It lies between structured and unstructured data.

- Does not have a fixed schema.
- Contains metadata or tags.
- Flexible structure.
- Easier to process than unstructured data.

Example (XML)

```
<Student>
  <Name>Amit</Name>
  <Marks>85</Marks>
</Student>
```

Advantages

- Flexible and adaptable.
- Easier to exchange between systems.
- Can store complex information.

Examples

- XML documents
- JSON files
- HTML pages
- Emails

Limitations

- More difficult to analyze than structured data.

- Requires special processing techniques.

3. Unstructured Data: refers to data that does not have a predefined format, schema, or organization. It cannot be easily stored in traditional relational databases. This is the most common type of data generated today.

- No fixed structure.
- Difficult to store and analyze using conventional methods.
- Exists in multiple formats.
- Requires advanced Big Data tools for processing.

- Documents
- Chat messages

Advantages

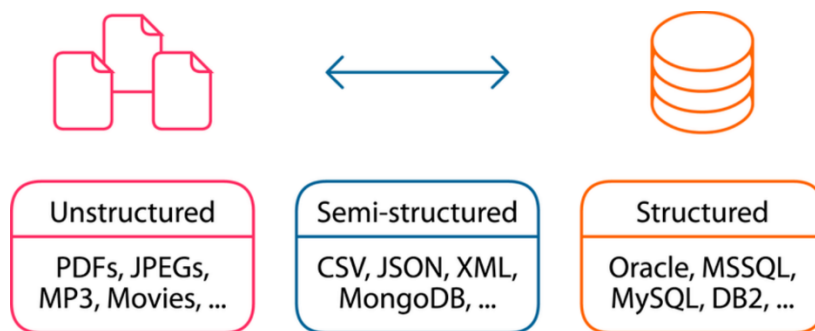
- Contains rich and valuable information.
- Provides deeper insights into user behavior and preferences.

Examples

- Images
- Videos
- Audio files
- Social media posts
- Emails

Limitations

- Difficult to manage.
- Requires large storage capacity.
- Complex analysis techniques are needed.



Importance of Unstructured Data

- **Largest Source of Data:** More than 80% of the world's data is unstructured, including social media content, images, videos, emails, and documents.
- **Better Customer Insights:** Organizations analyze reviews, comments, and social media posts to understand customer preferences and behavior.
- **Improved Decision Making:** Analysis of unstructured data helps businesses make informed decisions based on real-world trends and information.
- **Supports Advanced Analytics:** Technologies such as AI, Machine Learning (ML), and Natural Language Processing (NLP) rely heavily on unstructured data.
- **Healthcare Applications:** Medical images, doctor's notes, and patient records provide valuable information for diagnosis and treatment.
- **Fraud Detection and Security:** Emails, messages, and user activities can be analyzed to identify fraud and security threats.
- **Business Innovation:** Unstructured data helps organizations discover new opportunities, improve products, and enhance customer experiences.

Comparison of Data Types

Feature	Structured Data	Semi-Structured Data	Unstructured Data
Schema	Fixed	Partial	None
Organization	Rows and Columns	Tags and Metadata	No predefined structure
Storage	Relational Databases	XML, JSON Databases	Hadoop, NoSQL, Data Lakes
Processing	Easy	Moderate	Complex
Examples	Student Records, Banking Data	XML, JSON, Emails	Images, Videos, Social Media Posts

Applications of Big Data

Industry Examples of Big Data

1. **Banking:** Big Data helps banks detect fraud, assess credit risk, personalize financial services, and improve customer relationship management.
2. **Retail:** Retail companies analyze customer purchasing patterns, manage inventory, forecast demand, and provide personalized product recommendations.
3. **Telecommunications:** Telecom companies use Big Data to monitor network performance, predict equipment failures, reduce customer churn, and improve service quality.
4. **Transportation:** Transportation organizations analyze traffic patterns, optimize routes, manage fleets, and improve passenger safety and convenience.
5. **Manufacturing:** Manufacturers use Big Data for predictive maintenance, quality control, supply chain optimization, and production planning.

Web Analytics

Web Analytics is the process of collecting, measuring, analyzing, and reporting website data to understand user behavior and improve website performance. **Components**

- **Data Collection:** Gathering information about website visitors.
- **Data Measurement:** Measuring user activities and interactions.
- **Data Analysis:** Identifying patterns and trends.
- **Reporting:** Presenting insights through reports and dashboards.
- **Conversion Rate:** Percentage of visitors who complete a desired action such as registration or purchase.

Benefits

- **Improved User Experience:** Helps identify user preferences and website issues.
- **Better Marketing Decisions:** Measures effectiveness of marketing campaigns.
- **Increased Sales:** Optimizes website performance and conversions.
- **Customer Understanding:** Provides insights into visitor behavior.

Metrics

- **Page Views:** Total number of times web pages are viewed by users.
- **Visitors:** Number of unique users visiting the website.
- **Bounce Rate:** Percentage of visitors who leave the website after viewing only one page.

Big Data in Marketing

1. Big Data enables organizations to understand customers, predict behavior, and deliver personalized marketing strategies.
2. **Customer Segmentation:** Divides customers into groups based on demographics, interests, purchasing behavior, or location.
3. **Customer Behavior Analysis:** Analyzes customer activities and purchasing patterns to understand preferences and future needs.
4. **Personalized Marketing:** Delivers customized advertisements, offers, and recommendations based on customer interests.
5. **Recommendation Systems:** Suggests products or services based on previous purchases and browsing history.
6. **Sentiment Analysis:** Analyzes customer opinions from reviews, comments, and social media posts to determine positive or negative sentiments.
7. **Targeted Advertising:** Displays advertisements to specific customer groups that are most likely to be interested in the product.

Fraud Detection using Big Data

Types of Fraud

1. **Banking Fraud:** Illegal activities such as account manipulation, identity theft, and unauthorized transactions.
2. **Credit Card Fraud:** Unauthorized use of credit card information for financial gain.
3. **Insurance Fraud:** Submission of false insurance claims to obtain benefits illegally.
4. **Online Transaction Fraud:** Fraudulent activities involving online purchases, payments, or digital transactions.

Fraud Detection Process

1. Collect transaction and customer data.
2. Analyze patterns and behaviors.
3. Identify suspicious activities.
4. Generate alerts for investigation.
5. Take preventive actions.

Benefits

- **Real-Time Detection:** Identifies fraud immediately.
- **Reduced Financial Losses:** Prevents fraudulent transactions.
- **Improved Security:** Enhances protection of customer information.
- **Better Compliance:** Supports regulatory requirements.

Risk Analytics using Big Data

Risk Management: Risk management involves identifying, assessing, and controlling potential threats that may affect an organization.

Types of Risks

1. **Financial Risk:** Risk of financial loss due to market fluctuations, investments, or economic conditions.

2. **Operational Risk:** Risk arising from internal failures, system breakdowns, or human errors.
3. **Market Risk:** Risk caused by changes in market prices, interest rates, or exchange rates.

Role of Big Data in Risk Analysis

- **Risk Identification:** Detects potential risks using historical and real-time data.

- **Predictive Analysis:** Forecasts future risks and trends.
- **Better Decision Making:** Supports informed risk management strategies.
- **Continuous Monitoring:** Tracks risks in real time.

Credit Risk Management

1. **Credit Risk** Credit risk is the possibility that a borrower may fail to repay a loan or meet financial obligations.
2. **Credit Scoring:** A method of evaluating a customer's creditworthiness based on financial history and behavior.
3. **Loan Approval Analysis:** Big Data helps financial institutions assess loan applications by analyzing customer profiles and repayment capacity.

Data Sources Used

- Credit history
- Transaction records
- Income details
- Loan repayment records
- Customer behavior data

Benefits

- **Better Loan Decisions:** Improves approval accuracy.
- **Reduced Default Risk:** Identifies high-risk borrowers.
- **Faster Processing:** Automates risk assessment.
- **Improved Profitability:** Reduces financial losses.

Big Data and Algorithmic Trading

1. **Algorithmic Trading:** Algorithmic trading uses computer programs to automatically execute financial transactions based on predefined rules.
2. **Trading Algorithms:** Algorithms analyze market data and make trading decisions without human intervention.
3. **Market Data Analysis:** Big Data systems process large volumes of financial data, news feeds, and market trends to identify opportunities.
4. **Real-Time Decision Making:** Trading systems make instant decisions based on continuously changing market conditions.

Advantages

- **High Speed:** Executes trades within milliseconds.
- **Reduced Human Error:** Minimizes emotional decision-making.
- **Better Accuracy:** Uses data-driven strategies.
- **Efficient Market Analysis:** Processes huge datasets quickly.

Big Data in Healthcare

1. **Healthcare Data Sources:** Healthcare data is collected from hospitals, clinics, laboratories, wearable devices, and patient records.
2. **Electronic Health Records (EHR):** Digital versions of patient medical histories, diagnoses, treatments, and prescriptions.
3. **Patient Monitoring:** Wearable devices and sensors continuously monitor patient health conditions in real time.
4. **Disease Prediction:** Big Data helps predict diseases by analyzing medical

records, lifestyle data, and genetic information.

5. **Hospital Management:** Supports resource allocation, staff scheduling, inventory management, and operational efficiency.

Big Data in Medicine

1. **Personalized Medicine:** Medical treatments are customized according to an individual's genetic and health profile.
2. **Genomic Analysis:** Analysis of DNA and genetic information to identify disease risks and treatment options.

3. **Drug Discovery:** Big Data accelerates the identification and development of new medicines.

4. **Clinical Research:** Researchers analyze large datasets to evaluate treatments and improve medical knowledge.

5. **Precision Medicine:** A medical approach that provides targeted treatments based on patient-specific characteristics.

Advertising and Big Data

1. **Digital Advertising:** Uses online platforms and customer data to deliver advertisements through websites, social media, and mobile applications.
2. **Customer Profiling:** Creates detailed profiles of customers based on demographics, interests, and behavior.
3. **Ad Targeting:** Displays advertisements to users who are most likely to respond positively.
4. **Behavioral Advertising:** Uses browsing history, search activity, and online behavior to personalize advertisements.
5. **Ad Performance Analysis:** Measures advertisement effectiveness using metrics such as clicks, impressions, and conversion rates.

Big Data Technologies

Big Data technologies are tools and frameworks used to store, process, manage, and analyze large volumes of structured, semi-structured, and unstructured data. These technologies help organizations handle data that cannot be efficiently processed using traditional systems.

- 1) **Hadoop** An open-source framework that enables the distributed storage and processing of large datasets across multiple computers. It consists mainly of HDFS (storage) and MapReduce (processing).

Features

- Distributed data storage.
- Fault tolerance.
- Scalability.
- Cost-effective processing of large datasets.

Applications

- Data warehousing.

- Log analysis.
- Large-scale data processing.

- 2) **NoSQL Databases:** Non-relational databases designed to store and manage large volumes of diverse and rapidly changing data. **Features**

- Flexible schema.
- High scalability.
- Fast data access.
- Support for unstructured data.

Examples

- MongoDB
- Cassandra

- HBase
- CouchDB

Applications

- Social media platforms.
- Real-time web applications.
- Big Data analytics.

3) **Apache Spark:** A fast, open-source data processing framework that performs large-scale data analytics using in-memory computing. **Features**

- High-speed processing.
- Real-time analytics.
- In-memory computation.
- Support for machine learning and streaming.

Applications

- Data analytics.
- Machine learning.
- Real-time data processing.

4) **Hive:** A data warehouse tool built on Hadoop that allows users to query and analyze large datasets using Hive Query Language (HiveQL), which is similar to SQL.

Features of Hive

- SQL-like query language.
- Suitable for data warehousing.
- Easy data summarization.
- Runs on Hadoop.

Applications

- Data analysis.
- Report generation.
- Business intelligence.

5) **Pig:** A high-level data processing platform that uses a scripting language called Pig Latin to analyze large datasets on Hadoop.

Features

- Simple scripting language.
- Supports complex data processing.
- Reduces coding effort.
- Works on Hadoop.

Applications

- Data transformation.
- ETL (Extract, Transform, Load) operations.
- Data analysis.

6) **Kafka:** A distributed messaging and streaming platform used to collect, store, and process real-time data streams.

Features

- Real-time data streaming.
- High throughput.
- Fault tolerance.
- Scalable architecture.

Applications

- Log monitoring.
- Event tracking.
- Real-time analytics.

Technology	Main Purpose
Hadoop	Distributed storage and processing
NoSQL Databases	Storage of non-relational data
Apache Spark	Fast data processing and analytics
Hive	SQL-based data warehousing
Pig	Data processing using scripts
Kafka	Real-time data streaming

Unit 1: Big Data and Hadoop Ecosystem

1. Introduction to Big Data

- What is Big Data?
 - Characteristics of Big Data (Vs)
 - Volume
 - Velocity
 - Variety
 - Veracity
 - Value
 - Why Big Data?
-

2. Evolution of Big Data

- Convergence of Key Trends
 - Internet Growth
 - Social Media
 - Cloud Computing
 - Mobile Computing
 - IoT
 - Cheap Storage
-

3. Types of Data

- Structured Data
 - Semi-Structured Data
 - Unstructured Data
 - Importance of Unstructured Data
-

4. Industry Applications of Big Data

- Industry Examples of Big Data
 - Banking
 - Retail
 - Telecommunications
 - Transportation
 - Manufacturing
-

5. Web Analytics

- Definition
 - Components
 - Metrics
 - Page Views
 - Visitors
 - Bounce Rate
 - Conversion Rate
 - Benefits
-

6. Big Data in Marketing

- Customer Segmentation
- Customer Behavior Analysis

- Personalized Marketing
 - Recommendation Systems
 - Sentiment Analysis
 - Targeted Advertising
-

7. Fraud Detection using Big Data

- Types of Fraud
 - Banking Fraud
 - Credit Card Fraud
 - Insurance Fraud
 - Online Transaction Fraud
 - Fraud Detection Process
 - Benefits
-

8. Risk Analytics using Big Data

- Risk Management
 - Types of Risks
 - Financial Risk
 - Operational Risk
 - Market Risk
 - Role of Big Data in Risk Analysis
-

9. Credit Risk Management

- Credit Risk
 - Credit Scoring
 - Loan Approval Analysis
 - Data Sources Used
 - Benefits
-

10. Big Data and Algorithmic Trading

- Algorithmic Trading
 - Trading Algorithms
 - Market Data Analysis
 - Real-Time Decision Making
 - Advantages
-

11. Big Data in Healthcare

- Healthcare Data Sources
 - Electronic Health Records (EHR)
 - Patient Monitoring
 - Disease Prediction
 - Hospital Management
-

12. Big Data in Medicine

- Personalized Medicine

- Genomic Analysis
 - Drug Discovery
 - Clinical Research
 - Precision Medicine
-

13. Advertising and Big Data

- Digital Advertising
 - Customer Profiling
 - Ad Targeting
 - Behavioral Advertising
 - Ad Performance Analysis
-

14. Big Data Technologies

- Hadoop
 - NoSQL Databases
 - Apache Spark
 - Hive
 - Pig
 - Kafka
 - Other Big Data Tools
-

15. Introduction to Hadoop

- What is Hadoop?
- Need for Hadoop
- Hadoop Features
- Hadoop Ecosystem
- Advantages of Hadoop

Hadoop Core Components

- HDFS
 - MapReduce
 - YARN
 - Hadoop Common
-

16. Open Source Technologies

- Meaning of Open Source
 - Benefits
 - Examples
 - Hadoop
 - Hive
 - Pig
 - Spark
 - Cassandra
-

17. Cloud and Big Data

- Cloud Computing Basics

- Cloud Storage
- Cloud Analytics
- Benefits of Cloud for Big Data
- Big Data as a Service (BDaaS)

18. Mobile Business Intelligence (Mobile BI)

- Definition
- Mobile Analytics
- Real-Time Reporting
- Mobile Dashboards
- Advantages

19. Crowdsourcing Analytics

- Definition
- Data Collection from Large Groups
- Crowdsourced Data Sources
- Applications
- Benefits and Challenges

20. Inter and Trans Firewall Analytics

Inter-Firewall Analytics

- Analytics Across Multiple Networks
- Security Monitoring
- Data Sharing Between Organizations

Trans-Firewall Analytics

- Analytics Beyond Organizational Boundaries
- Cross-Enterprise Data Analysis
- Security and Privacy Concerns

Exam Priority (Most Important)

Very Important:

1. Big Data & 5Vs
2. Why Big Data
3. Structured vs Semi-Structured vs Unstructured Data
4. Hadoop Introduction
5. Hadoop Components (HDFS, MapReduce, YARN)
6. Big Data Technologies
7. Big Data in Healthcare
8. Big Data in Marketing
9. Web Analytics
10. Fraud Detection & Risk Analytics

Moderately

11. Credit Risk
12. Algorithmic

Important:
Management
Trading

- | | | | | |
|---------|------------------------------------|---------------|--------|--------------|
| 13. | Advertising | and | Big | Data |
| 14. | Cloud and Big Data | | | |
| Usually | | Short | | Notes: |
| 15. | | Mobile | | BI |
| 16. | | Crowdsourcing | | Analytics |
| 17. | Open | | Source | Technologies |
| 18. | Inter and Trans Firewall Analytics | | | |